Haichao Sha

Renmin University of China
Research: Differential Privacy, DP Optimization

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Education

- **Nanyang Technological University** — Singapore
  Visiting student of CCDS — *Dec. 2025 – now*
- **Renmin University of China** — Beijing, China
  PhD student of Big Data Science and Engineering; GPA: 3.91/4.00 — *Sep. 2022 – now*
- **Hohai University** — Nanjing, China
  Bachelor of Computer Science; GPA: 4.78/5.00 (rank: 5/240) — *Sep. 2018 – June. 2022*

Publications & Research

- **Differentially Private Visual Learning with Public Subspace Augmented by Synthetic Data**
  ACM Multimedia 2025 Outstanding Paper Award (Top 1%) (CCF-A)
  - **SAS-DPSGD Framework**: Proposed Synthesis-Augmented Subspace DP-SGD (SAS-DPSGD), integrating public and synthetic data to enhance subspace diversity for private visual learning.
  - **Early Projection Mechanism**: Introduced an early projection strategy to reduce clipping bias, achieving faster convergence with significantly less synthetic data.
  - **Optimization Analysis**: Provided quantitative analysis of privacy-utility trade-offs and revealed saturation effects when increasing synthetic data in private optimization.
- **Differentially Private Stochastic Gradient Descent: Methodology, Theory, and Application**
  Chinese Journal of Computers (CCF-A Chinese)
  - **Methodology**: Established a three-level pipeline-oriented taxonomy of DP-SGD techniques for sampling strategies and privacy amplification, gradient clipping mechanisms and bias–variance trade-offs, noise injection and privacy accounting methods.
  - **Theory**: Provided a systematic and unified theoretical analysis of DP-SGD, clarifying convergence behavior, stability properties, and privacy-induced optimization effects.
  - **Application**: Analyzed DP-SGD in large language models, highlighting optimization bottlenecks and emerging privacy risks in training and fine-tuning.
- **Clip Body and Tail Separately: High-Probability Guarantees for DPSGD with Heavy Tails**
  Paper under Review
  - **DC-DPSGD Framework**: Identified the limitations of standard DP-SGD under heavy-tailed gradient noise, where conventional clipping becomes ineffective.
  - **Tail-Aware Clipping Mechanism**: Proposed a tail-aware discriminative clipping mechanism that privately classifies gradients into body and tail via subspace identification and applies separate clipping thresholds to jointly reduce clipping bias and variance.
  - **Optimization Analysis**: Derived tighter high-probability optimization guarantees for DP-SGD under heavy-tailed assumptions.

  We are currently conducting research on privacy auditing of large language models and zero-order differential privacy optimization.

Projects

- **National Natural Science Foundation of China (NSFC) – General Program**: Participated in a general NSFC project on privacy-preserving techniques in machine learning.
- **National Natural Science Foundation of China (NSFC) – Joint Funds**: Participated in a key joint NSFC project on data security management and privacy computing for lakehouse architectures.
- **CCF–Tencent Rhino-Bird Special Fund**: Participated in a CCF-Tencent Special Fund project (PI: Prof. Yuncheng Wu), focusing on privacy computing on data analysis.

Honors and Awards

- Outstanding Paper, ACM MM, 2025
- Outstanding Graduate, 2022
- Excellent Undergraduate Thesis, 2022
- Challenge Cup National Undergraduate Academic and Technological Competition (Jiangsu Province) Second Prize, 2021
- National Scholarship for Undergraduate Students (China) National Scholarship, 2020
- Jiangsu Provincial Mathematics Competition (Group A) First Prize, 2019
- National College Mathematics Competition (Jiangsu Division) Second Prize, 2019
- China Undergraduate Computer Design Competition Second Prize, 2020
- Jiangsu Undergraduate Computer Design Competition Third Prize, 2020
- National Undergraduate Mathematical Contest in Modeling (Jiangsu Province) Third Prize, 2020